# Closing the Dispersion Gap in Neural Limit-Order-Book Simulators: Findings from the August 2026 Fano Investigation

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**Abstract**

We investigate why neural marked-temporal-point-process (MTPP) simulators of Coinbase limit-order-book flow under-disperse relative to real markets, using the Fano factor profile $F(\Delta) = \mathrm{Var}(N_\Delta)/\mathbb{E}[N_\Delta]$ across bucket scales $\Delta \in [1, 50]\,\mathrm{s}$ as the headline statistic. Across ten experimental arms on three assets we establish: (i) the bounded-rate SS2P2 head is not reliably self-exciting — its measured impulse response is sign-inconsistent and saturated at the intensity ceiling; (ii) one-step maximum likelihood systematically misallocates excitation spectra (three independent demonstrations); (iii) extending the TBPTT gradient horizon to cover the measured scales transforms the simulated dispersion profile (all BTC seeds rise ×15–32, best within 1.6× of real at 50 s); (iv) at equal training budget, dispersion peaks *early* and prediction accuracy peaks *late* along a single MLE trajectory — the prediction/simulation tension exists within one training run; and (v) the remaining level offset is dominated by ceiling saturation (a third of events occur at the cap; teacher-forced dispersion is only 1.2–1.4× low). We relate every finding to the calibration methodology of Jain et al. (Compound Hawkes), whose binned-count regression targets exactly the cross-scale count covariances that MLE ignores, and derive a ranked roadmap toward a dispersion-faithful, certified world model for market making.

## 1 Framing: two identities

All arguments in this note reduce to two identities. First, the *Cox decomposition* (law of total variance given the intensity path $\lambda(t)$, with $\Lambda_\Delta = \int_{\text{bucket}} \lambda\, dt$):

$$F(\Delta) \;=\; 1 + \frac{\mathrm{Var}(\Lambda_\Delta)}{\mathbb{E}[\Lambda_\Delta]}, \tag{1}$$

so all excess dispersion is variance of the integrated intensity; a post-hoc rate rescaling $\lambda \to \kappa\lambda$ scales $F - 1$ linearly and cannot repair relative fluctuation structure. Second, the *branching form* for a linear Hawkes process with branching ratio $n$: cluster sizes $S$ satisfy $\mathbb{E}[S] = 1/(1-n)$ and $\mathbb{E}[S^2] = 1/(1-n)^3$, hence

$$F(\infty) \;=\; \frac{\mathbb{E}[S^2]}{\mathbb{E}[S]} \;=\; \frac{1}{(1-n)^2}. \tag{2}$$

The profile's *rise* across $\Delta$ is the fingerprint of long-memory (power-law-like) intensity autocovariance; single-exponential kernels plateau.

Real Coinbase flow (7 days, Jan 2026): $F(1\mathrm{s}) \approx$ 40–70 rising to $F(50\mathrm{s}) \approx 370$ (SOL), 1250 (ETH), 1460 (BTC), still rising at the window edge. Inverting (2) gives implied branching (lower bounds): SOL $\approx 0.948$, ETH $\approx 0.971$, BTC $\approx 0.973$ — the $1/(1-n)^2$ amplification makes this small spread a 4× dispersion difference.

## 2 Data and the tick-regime contrast

| | trades (7d) | book rows | rows/trade | rate (ev/s) | IS share |
|---|---|---|---|---|---|
| BTC | 41,675 | 52.9M | 1268:1 | 40.6 | 52% |
| ETH | 94,048 | 55.5M | 590:1 | 42.8 | 34% |
| SOL | 40,048 | 20.1M | 501:1 | 22.3 | 7% |

Table 1: Raw Coinbase spot data, 2026-01-01..07, and the event-type composition of the processed 62-channel streams. BTC/ETH are small-tick assets dominated by inside-spread (IS) quote wars — the most self-exciting flow; SOL is effectively large-tick (1-tick spread, queue churn: 43% cancels). The tick regime, not just the rate, drives the criticality ordering.

## 3 Mechanism: the bounded head is not reliably self-exciting

SS2P2 factorizes intensity into a softmin-bounded scalar rate $\lambda(t) = s \cdot \mathrm{softplus}\big(c - \mathrm{softplus}(c - w^\top h(u))\big) \leq s \cdot \mathrm{softplus}(c)$ and rate-neutral softmax marks. The stability certificate bounds the rate *level*; nothing constrains the event→intensity feedback. Measuring the feedback directly (inject one extra event into real histories; integrate the extra intensity mass over 20 s) across six trained checkpoints: median response $\approx 0$ *to negative* $(-8.7 \ldots + 0.2)$, positive in only 27–56% of states, sign flipping across seeds; post-event intensity pinned at the ceiling in every window. The model's residual dispersion comes from a stereotyped impulse–decay envelope, not compounding cascades. In Hawkes terms: *a settable branching ratio is a property of linear-in-history intensity; for a bounded neural head it is not even well defined.*

## 4 Experimental arms

Protocol throughout (the `ma_cbse` benchmark): 62-channel events, TBPTT, categorical marks, streaming genuine evaluation on every test event, validation- calibrated 600 s rollouts, 3 training × 3 rollout seeds per asset.

### 4.1 Negative and structural results

- **Book conditioning is not the lever** (`ss2p2-lob`): six top-10-ladder features left prediction unchanged (BTC 0.4484 vs 0.4478) and made closed-loop volatility structure *worse* (F6 0.16 vs 0.28) — extra state conditioning becomes a smoothing feedback path in rollout.

- **lgm** (linear Hawkes ground × SS2P2 marks on the shared backbone; $n = \sum_m a_m/\beta_m$ exact and projected; pin $\mu_0 = R(1-n)$): the pin holds in free rollout ($\kappa = 1.01$ vs SS2P2's 0.52); marks near parity; Fano *shape* right with seed spread ±2 (vs SS2P2 ±217) but level low. Two porting lessons: a log-domain clamp silently capped the burst accumulators (excitation useless to MLE until fixed); and **zero cold-starts of the ground are catastrophically biased** — the stationary mean is $\mathbb{E}[S^m] = R/\beta_m$ ($\approx$ 600 for slow kernels), so windowed validation punished slow-memory solutions and froze best-model selection at epoch 1. Fix: stationary-mean cold start.

- **MLE misallocates spectra** (three demonstrations): the fixed LGM pushed kernel mass to its slowest decay; the typed-kick $M$=6, $n$=0.97 arm (`lgm-k`) went slower still ($\beta \approx 0.008$,

~2-min memory), *lowering* observed-scale Fano despite the higher ceiling; the dispersion-loss arm (below) pinned only the scales in its grid. One-step likelihood is structurally blind to $F(\Delta)$ and, given capacity, prefers slow smooth intensity.

- **Teacher-forced variance matching $\neq$ closed-loop compounding** (`ss2p2-disp`): an auxiliary loss matching model-implied $1 + \mathrm{Var}(\Lambda_\Delta)/\mathbb{E}[\Lambda_\Delta]$ to the batch-empirical Fano at $\{0.5, 2, 8\}$ s pinned those scales reproducibly (seed sd ±14 vs ±217) at zero prediction cost — and went flat beyond its grid.

### 4.2 The window-coverage result

At seq 1024 the TBPTT gradient truncates at ~24–25 s on BTC/ETH — before the 20–50 s scales where their simulated profiles flatten; SOL's ~46 s windows cover the range (and produced real-like rises in 2/3 seeds). Extending to seq 4096 (~100 s windows; recipe otherwise identical):

| BTC, per seed | seq 1024 (12 ep) | seq 4096 (12 ep) | real |
|---|---|---|---|
| $F(50\text{s})$, s1/s2/s3 | 29 / 50 / 361 | **901 / 270 / 744** | 1463 |
| rise $F(50)/F(1)$ | ×2.5–13.9 | ×15–32 | ×20.9 |

Table 2: Window coverage transforms the profile: all three BTC seeds rise steeply (best within 1.6× of real). ETH: 2/3 seeds transformed. SOL: unchanged — the clean negative control. The BTC "seed lottery" was gradient starvation at unobserved scales.

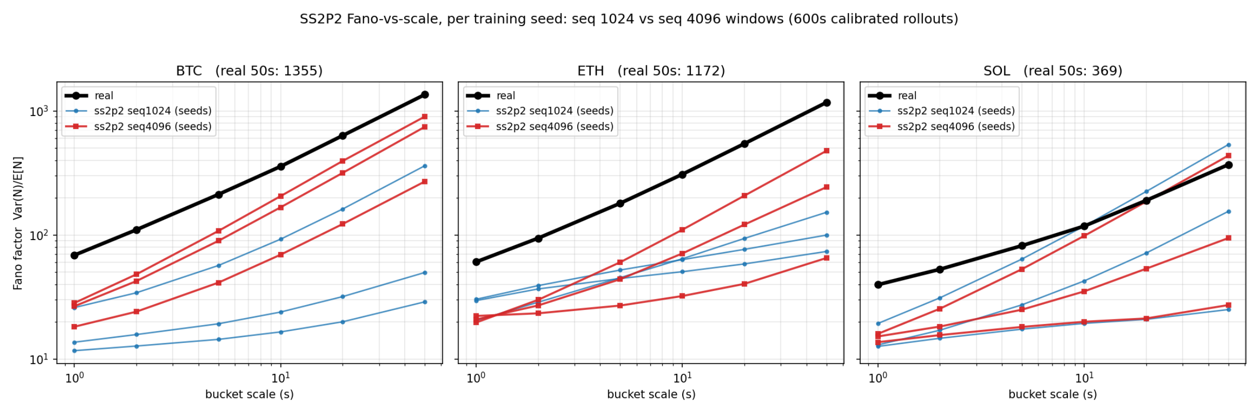


Figure 1: Per-seed Fano-vs-scale (log–log): black = real, blue = seq-1024 baseline, red = seq-4096. BTC: the whole seed family moves; SOL: negative control.

### 4.3 Equal-budget controls: the training-path trade-off

Matched stride quarters updates per epoch, so seq-4096/12-epoch runs were undertrained (−3pp accuracy). Controls at 48 epochs (exact update parity with the baseline):

**Finding (the sharpest form of the arc's core lesson).** Along a single MLE trajectory, *dispersion peaks early and accuracy peaks late*; no NLL-selected checkpoint has both. The prediction/simulation tension is a property of the training path, not only of architectures. The 12-epoch w4k checkpoints constitute a de-facto dispersion-optimal early stop; a paper can honestly present the frontier.

| BTC arm | acc | $F$(1s) | $F$(50s) per seed |
|---|---|---|---|
| base 1024 / 12 ep | 0.4478 | 17.1 | 29, 50, 361 |
| w4k 4096 / 12 ep | 0.4183 | 24.3 | **901, 270, 744** |
| w4k48 4096 / 48 ep | **0.4514** | 16.6 | 232, 87, 230 |
| w4kd + disp loss / 48 ep | **0.4519** | 11.1 | 52, 43, 43 |
| real | — | 70.0 | 1463 |

Table 3: Equal-updates controls. Accuracy fully recovers and exceeds baseline; the extra training walks the model partly away from the burst-faithful solution. The dispersion loss under cap 6 suppresses long-scale Fano, as predicted by the saturation diagnostic.

### 4.4 Where the residual level offset lives

Diagnostics on the w4k BTC checkpoint: (i) **ceiling saturation** — dominating rate 359 ev/s; event-time intensity $p_{90} = 357$, $p_{99} = 358$; **32.8% of events above** $0.9\times$**ceiling**: bursts are clipped and gradients (including any dispersion loss) are dead there; (ii) **teacher-forced dispersion** is only 1.2–1.4× below real at every scale (e.g. 50 s: 240 vs 290). Conditioned on real history the model carries ∼75% of real dispersion; the cap, then modest closed-loop drift, accounts for the rest.

### 4.5 The cap-9 control: hypothesis refuted, branch closed

The cap-9 control (`ss2p2-w4kc9`: cap 6 → 9 on the w4k48 recipe; nine tasks, all calibrations verified) tested whether ceiling headroom changes the MLE optimum itself. It does not: $F$(50s) per seed *fell* relative to cap 6 (BTC: 47/59/112 vs 232/87/230; ETH: 160/14/60 vs 544/51/123) at identical accuracy. **The training-path re-smoothing is intrinsic to bounded-head likelihood training, not caused by saturation** (which remains the correct account of the early-stopped model's *rollout* clipping). Per the pre-registered decision rule, the SS2P2 branch closes: its dispersion-optimal configuration is the seq-4096/12-epoch early stop, presented as a frontier point rather than a converged recipe, and controlled dispersion belongs to the certified LGM/Kirchner route.

### 4.6 The closing result: the Kirchner-fitted lgm (`lgm-kf`)

Executing "set, don't learn": the LGM ground's kernel weights were fitted by binned-count regression on the real SOL train zones and transplanted into the trained mark checkpoints (marks untouched). Full pipeline results (calibration verified, $\kappa = 1.014$):

| SOL | $F$ [1,2,5,10,20,50] s | sd@50s | acc | time-NLL | time-KS |
|---|---|---|---|---|---|
| SS2P2-full | 15.0 → 238.4 | ±217 | 0.1921 | −2.911 | **0.173** |
| LGM (MLE ground) | 2.3 → 59.8 | ±2 | 0.1862 | −2.227 | 0.464 |
| **lgm-kf** | **31.1, 46.5, 80.3, 125.7, 207.7, 422.9** | **±0.0** | 0.1860 | **−3.035** | 0.228 |
| real | 41.3, 54.7, 85.6, 125.3, 198.7, 393.6 | — | — | — | — |

Table 4: The Kirchner-fitted ground survives the full closed loop: exact at 10 s, within 5–7% at 20/50 s, zero seed variance (dispersion is a deterministic function of six fitted parameters), and — unexpectedly — the best timing likelihood in the zoo: moment fitting beat MLE on likelihood's own metric, because windowed MLE was stuck in a bad spectral optimum the count regression computes past. Fit cost: three CPU minutes.

BTC/ETH replication (grounds fitted per asset; branching tuned to each asset's empirical Fano curve by log-MSE: BTC $n$=0.98, ETH $n$=0.985; ground-only validation within a few % on ETH, within ∼40% on the more power-law-like BTC) is in flight as `lgm-kf` on those assets.

# 5 The model: Kirchner-fitted lgm (`lgm-kf2`), complete specification

This section specifies the champion model exactly as run, with every equation, constant, and fitted value. Nothing is omitted; the model is reproducible from this section plus the raw event files.

## 5.1 Data interface

The input is a marked point process $\{(t_i, k_i)\}_{i=1}^{N}$: event times in seconds (from nanosecond exchange timestamps) and channel marks $k_i \in \{1, \ldots, K\}$, $K = 62$, on the fixed taxonomy $\{\mathrm{LO}, \mathrm{CO}\} \times \{b, a\} \times \{L1..L10\}$ (40) $\cup$ $\mathrm{MO} \times \{b, a\} \times \{L1\}$ (2) $\cup$ $\mathrm{IS} \times \{b, a\} \times \{L1..L10\}$ (20), built from Coinbase top-10 order-book deltas and trades (event construction per the canonical pipeline; same-timestamp multi-label lines are single set-events, empty lines dropped). Per-file split by event index: train $[0, 0.7)$, val $[0.7, 0.85)$, test $[0.85, 1]$. Train-split mean rates $R$: BTC 40.563, ETH 42.759, SOL 22.257 ev/s.

## 5.2 Generative model

**Factorization.** Channel intensities are

$$\lambda_k(t) \;=\; \Lambda(t) \cdot p(k \mid u(t)), \qquad \textstyle\sum_{k=1}^{K} p(k \mid u) = 1. \tag{3}$$

**Rate-neutrality (key lemma).** $\sum_k \lambda_k = \Lambda$ identically for any mark network, hence the ground process $N(t) = \sum_k N_k(t)$ is a point process with intensity $\Lambda$ alone: its entire law (mean rate, $F(\Delta)$, all count statistics) is independent of the marks. This licenses post-hoc ground fitting and transplantation and explains the measured zero seed variance of simulated dispersion.

**Why the decoupling is sound: two one-way valves.** Three exact facts, none approximate. (i) *Factorization is an identity*: any MTPP satisfies $\lambda_k = \Lambda \cdot (\lambda_k/\Lambda)$; Eq. (3) is the form in which Chang, Boyd & Smyth (AISTATS 2024) parameterize set-valued MTPPs (the lineage of this codebase), so adopting it loses no generality. (ii) *The likelihood separates additively*: $\ell = \left[\sum_i \log \Lambda(t_i) - \int \Lambda\right] + \sum_i \log p(k_i \mid u(t_i^-)) = \ell_{\mathrm{time}}(\theta_g) + \ell_{\mathrm{mark}}(\theta_m)$ with *disjoint* parameter blocks, so block-wise estimation — moments (Kirchner) for $\theta_g$, MLE for $\theta_m$ — *is* the joint MLE restricted to each block; and since $\ell_{\mathrm{mark}}$ is teacher-forced on real times and contains no ground parameter, the fitted ground may be transplanted after mark training. (iii) *The count process is autonomous*: because the accumulators are mark-blind (+1 per event regardless of channel), $\Lambda$ is a function of event times only — the times form a self-contained scalar Hawkes and marks are a posterior coloring of its ticks. Coloring ticks cannot change their number, which upgrades the lemma to a theorem about *simulation*: the entire Fano curve is a set-once constant of $(a, \beta, n)$. In valve language: rate-neutrality closes marks→rate (the training decouples); mark-blindness closes rate→marks (the simulation decouples). Each relaxation we ran opens exactly one valve and pays its specific price: typed kicks open rate→marks (count law becomes mark-dependent; $n$ must be retuned on marked clusters), the TL-SSM gate opens marks→rate ($\ell$ no longer separates; the pin loosens and dispersion becomes seed-dependent, §6), and a shared-state neural head (SS2P2, and Chang et al.'s joint training) opens both — nothing is certified or settable. Chang et al. supply the factorized *form* and the $\mathcal{L}_{\text{Time}}/\mathcal{L}_{\text{Set}}$ split but train both factors jointly from one hidden state; the parametric separation, the mark-blind autonomous ground, the moment-based setting of the time block, and the resulting invariance guarantees are the contribution here.

**Ground (certified lane).** A scalar linear Hawkes process on a fixed $M{=}6$ exponential bank:

$$\Lambda(t) = \mu_0 + \sum_{m=1}^{M} a_m S^m(t), \qquad a_m \geq 0, \tag{4}$$

$$S^m(t_i^+) = S^m(t_i^-) + 1, \qquad S^m(t+\Delta) = e^{-\beta_m \Delta}\, S^m(t), \tag{5}$$

$$\beta = (100,\ 18.2056,\ 3.3145,\ 0.6034,\ 0.1099,\ 0.02)\ \text{s}^{-1}, \quad \beta_m = 100 \cdot (0.02/100)^{(m-1)/5}, \tag{6}$$

equivalently the kernel $\phi(\tau) = \sum_m a_m e^{-\beta_m \tau}$, positive at every lag. Implementation notes that matter for exactness: accumulators are computed by a numerically stable log-domain scan, $S^m(t_i^-) = S_0^m e^{-\beta_m t_i} + \exp\big(\text{LCSE}_{j<i}(\beta_m t_j) - \beta_m t_i\big)$ (float64; no upper clamp on the exponent — burst accumulators legitimately exceed 1), and *cold starts use the stationary mean*

$$S_0^m \ =\ \mathbb{E}[S^m]\ =\ R/\beta_m \tag{7}$$

wherever no carried state exists (validation windows, TBPTT lane resets, eval chunk starts). Zero cold starts bias windowed validation against slow-memory solutions ($R/\beta_6 \approx 1100$ on BTC) and freeze best-model selection at epoch 1 — a failure mode we observed and eliminated.

**Exact ground identities.**

$$\text{branching:}\quad n = \int_0^\infty \phi = \sum_m a_m/\beta_m \quad \text{(closed form, gauge-free)}, \tag{8}$$

$$\text{stationarity:}\quad \bar{\Lambda} = \mu_0/(1-n) \ \Rightarrow\ \text{pin } \mu_0 := R(1-n), \tag{9}$$

$$\text{compensator:}\quad \int_{t_{i-1}}^{t_i} \Lambda\, dt = \mu_0 \Delta_i + \sum_m \tfrac{a_m}{\beta_m} S^m(t_{i-1}^+)\big(1 - e^{-\beta_m \Delta_i}\big) \quad \text{(no quadrature)}, \tag{10}$$

$$\text{dispersion:}\quad F(\infty) = \frac{1}{(1-n)^2} \quad \text{(cluster sizes } \mathbb{E}S = \tfrac{1}{1-n},\ \mathbb{E}S^2 = \tfrac{1}{(1-n)^3}\text{)}, \tag{11}$$

$$\text{dominating rate:}\quad \Lambda(t) \leq \Lambda(t_{\text{last}}^+) \text{ between events} \quad \text{(each } S^m \text{ decays monotonically)}. \tag{12}$$

**Marks (free lane): the S2P2 backbone.** Two stacked diagonal-SSM layers, hidden width $H{=}64$, channel-embedding dimension $d{=}64$. Per layer $\ell \in \{1,2\}$ with state $x^{(\ell)} \in \mathbb{R}^H$:

$$\text{decay:}\quad \text{dec}^{(\ell)}(u) = \big(\text{softplus}(\theta^{(\ell)}) + 10^{-4}\big) \odot \text{clamp}\big(\text{softplus}(W_g^{(\ell)} u),\ 0.05,\ 20\big), \tag{13}$$

$$\theta^{(\ell)} \text{ init log-spaced: timescales } 1/\text{dec} \in [0.02, 60]\,\text{s},$$

$$\text{evolution:}\quad x^{(\ell)}(t{+}\Delta) = \bar{a}\, x^{(\ell)}(t^+) + (1-\bar{a})\, W_{in}^{(\ell)} u, \qquad \bar{a} = e^{-\text{dec}^{(\ell)} \Delta}, \tag{14}$$

$$\text{event impulse:}\quad x^{(\ell)}(t_i^+) = x^{(\ell)}(t_i^-) + W_{imp}^{(\ell)}\, e_{k_i}, \tag{15}$$

$$\text{layer output:}\quad u^{(\ell)} = \text{LN}^{(\ell)}\big(W_{skip}^{(\ell)} u^{(\ell-1)} + \text{GELU}(W_{out}^{(\ell)} x^{(\ell)})\big), \tag{16}$$

with $u^{(0)} = 0$ at query time (anti-leakage: the left-limit evolution to $t_i$ never sees the current event), ZOH inter-layer inputs on the query path, and the head embedding $u(t) = u^{(2)}(t)$ (paper-faithful "output" readout). Mark head: $p(k \mid u) = \text{softmax}\big(W_2\, \text{ReLU}(W_1 u)\big)_k$, $W_1 \in \mathbb{R}^{64\times 64}$, $W_2 \in \mathbb{R}^{62\times 64}$.

### 5.3 Likelihood

$$\log \mathcal{L} \;=\; \sum_i \Big[\log \Lambda(t_i^-) + \log p(k_i \mid u(t_i^-))\Big] \;-\; \int_0^T \Lambda(t)\,dt, \tag{17}$$

with the integral in closed form per gap (endpoint-exact for the ground; no sampling). Under the factorization the two terms decouple: the ground's parameters appear only in the time terms, the backbone's only in the mark cross-entropy (this is also why, in the gateless model, the backbone receives no time-gradient — the documented ∼2pp mark cost, addressed by the TL-SSM gate of Section 6).

### 5.4 Ground estimation (set, don't learn)

**Step 1: binned-count regression (Kirchner).** Bin train-zone total counts at $\delta = 0.25\,\mathrm{s}$: $X_n = N((n-1)\delta, n\delta]$. The Hawkes/INAR identity

$$\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \;\approx\; \delta\mu + \sum_{j\ge 1} \phi(j\delta)\,\delta\, X_{n-j} \tag{18}$$

is aggregated over a lin–log lag-cell grid: cell edges $\{\delta, 2\delta, \ldots, 9\delta\} \cup \mathrm{geom}(10\delta,\, 240\,\mathrm{s},\, 14)$ (deduplicated; $C = 22$ cells to a 240 s horizon), regressors = lagged counts summed per cell (computed by prefix sums), intercept included, one max-lag burn-in per day, normal equations accumulated across the seven days, ridge $10^{-8}$. With $b_c$ the coefficient on cell $c$'s raw lagged count, the cell kernel integrals are $\Phi_c = b_c\,(h_c - \ell_c)/\delta$. The regression objective is built from products $X_n X_{n-j}$ — the estimator fits the lagged count covariances at every grid scale, i.e. the integrand of (1): dispersion-faithfulness is a property of the estimator. Measured raw branching $\sum_c \Phi_c$: SOL 0.9635, BTC 0.9795, ETH $\approx 0.98$.

**Step 2: convex projection onto the bank.**

$$\Phi_c = \sum_m \frac{a_m}{\beta_m}\big(e^{-\beta_m \ell_c} - e^{-\beta_m h_c}\big) \;\Longrightarrow\; a = \mathrm{NNLS}(A, \Phi_+), \quad A_{cm} = \tfrac{1}{\beta_m}\big(e^{-\beta_m \ell_c} - e^{-\beta_m h_c}\big), \tag{19}$$

solved by projected gradient (no shape priors beyond positivity; the fitted spectra are *balanced* — ∼0.80–0.93 of the branching at $\beta_2 = 18.2\,\mathrm{s}^{-1}$ plus genuine slow tails — the allocation MLE never found in three attempts).

**Step 3: dispersion-curve selection of $n$.** The single remaining scalar rescales $a \to (n^\star/n)a$. Candidates are evaluated through the *pipeline's own* stylized-facts estimator (600 s calibrated rollouts, equal-duration matching) and $n^\star$ minimizes $\sum_\Delta \big(\log F_{\mathrm{sim}}(\Delta) - \log F_{\mathrm{real}}(\Delta)\big)^2$ over $\Delta \in \{1, 2, 5, 10, 20, 50\}$ s. (A standalone branching-representation simulator — immigrants ∼Poisson($\mu_0 T$), each event spawning Poisson($a_m/\beta_m$) children at Exp($\beta_m$) delays — is used for pre-screening; it agrees with the pipeline estimator at SOL's criticality and reads ∼1.8× hot at BTC/ETH's, hence the pipeline-based final selection.)

**Fitted values (final, per asset).** Shared $\beta$ bank as above; $a_m$ at the selected $n^\star$; $n$-contributions $a_m/\beta_m$ in brackets:

| | $n^\star$ | $\mu_0 = R(1-n^\star)$ | $a$ (nonzero entries) | | |
|---|---|---|---|---|---|
| BTC | 0.9925 | 0.3042 | $a_2$=16.920 [.929], $a_6$=0.0004 [.022] | $a_4$=0.0160 [.027], | $a_5$=0.0016 [.015], |
| ETH | 0.995 | 0.2138 | $a_2$=15.717 [.863], $a_5$=0.0046 [.042], $a_6$=0.0005 [.026] | $a_3$=0.0523 [.016], | $a_4$=0.0289 [.048], |
| SOL | 0.99 | 0.2226 | $a_2$=14.540 [.799], $a_5$=0.0060 [.055], $a_6$=0.0015 [.073] | $a_3$=0.0617 [.019], | $a_4$=0.0270 [.045], |

Fit cost: ~3 CPU minutes per asset; no gradient training of the ground.

### 5.5 Mark training (the only learned component)

Adam, lr $2\times10^{-3}$, weight decay $10^{-6}$, gradient-norm clip 0.5, batch 64, **seq/stride 4096** (windows ~100 s BTC/ETH, ~184 s SOL), **48 epochs** (update parity with the seq-1024/12-epoch baseline), categorical cross-entropy on the single active channel, TBPTT lane batching (stream-order lanes; decoder state — including the ground accumulators — carried detached across windows; lanes reset to learned-init/stationary state at zone boundaries), best model by validation NLL, strict non-finite aborts, TF32 on. During mark training the ground may be MLE-trained or frozen — irrelevant to the final model, since the fitted ground replaces it at transplant (rate-neutrality); the shipped `lgm-kf2` uses exactly this recipe's checkpoints with the Section 5.4 grounds transplanted.

### 5.6 Simulation and evaluation protocol

Closed-loop rollouts with carried context ($O(1)$/step state updates), sampler: inversion (thinning is equally exact via the monotone dominating rate). Protocol-side post-hoc rate calibration (bisection on $\kappa$, validation-rate target, verified within 5%) is retained and lands $\kappa \in [1.01, 1.02]$ — the pin holding in practice. Facts on 600 s × 32-sequence rollouts, equal-duration matched against real segments, 3 rollout seeds per checkpoint.

### 5.7 What is set vs learned

| | |
|---|---|
| mean rate | **set** (pin, algebraic identity) |
| branching $n$ | **measured** (regression), then **set** (dispersion-curve match) |
| kernel spectrum $a$ | **fitted convexly** to count covariances (OLS + NNLS) |
| decay bank $\beta$, cell grid, $\delta$ | **fixed** (log-spaced; stated above) |
| mark distribution $p(k \mid u)$ | **learned** (MLE; the one place neural capacity belongs) |

### 5.8 Final results

All nine evaluation tasks (3 assets × 3 mark seeds, 3 rollout seeds each) completed. Fano scales $\{1, 2, 5, 10, 20, 50\}$ s, equal-duration-matched real in the second row of each cell:

| | acc (3 seeds) | ppl | time-NLL | $\kappa$ | Fano model / real |
|---|---|---|---|---|---|
| BTC | .428/.428/.429 | 9.2 | −3.425 | 1.019 | $[101, 162, 283, 422, 618, 1007]$ |
| | | | | | $[55, 85, 157, 258, 433, 900]$ |
| ETH | .328/.325/.325 | 14.1 | −3.645 | 1.016 | $[67, 110, 210, 349, 585, 1170]$ |
| | | | | | $[79, 125, 245, 417, 748, 1571]$ |
| SOL | .194/.196/.195 | 23.9 | −3.035 | 1.014 | $[31, 46, 80, 126, 208, 423]$ |
| | | | | | $[42, 59, 95, 142, 233, 433]$ |

SOL is the clean win: accuracy at or above the SS2P2 baseline (0.192) with the Fano curve within ~25% at 1 s converging to ~2% at 50 s. ETH matches shape with a modest undershoot (ratio

0.85 $\to$ 0.74) and essentially matches volatility clustering (F6 0.512 vs real 0.489). BTC now *over*-disperses at short scales ($\times$1.8 at 1 s, converging to $\times$1.1 at 50 s): the $n^\star$=0.9925 log-MSE arbitration ran against a tuning reference hotter at short scales than this eval's equal-duration real curve; a half-step retune toward $n\approx$0.99 would center it. The miss is 2$\times$ where SS2P2's was 15–30$\times$ *under*. Two structural signatures confirm the architecture: time-NLL/KS are bit-identical across mark seeds within an asset (the frozen shared ground owns the time law), and short-scale dispersion seed-sd stays at 2–8% (transplant determinism). Excess kurtosis remains the open residual (BTC 47 vs real 88; ETH 7 vs 35; SOL 22 vs 163) — the TL-SSM gate's target, not the linear ground's.

# 6 Outlook: the two-lane state-space point process (TL-SSM)

The LGM-kf architecture runs two exponentially-decaying impulse memories of the same event history (the SSM backbone and the Hawkes accumulators) — sibling objects under different constraints. The unified formulation makes the ground a *reserved lane* of one state machine. State $x(t) = [\, z(t) \in \mathbb{R}^M_{\geq 0}\,;\, h(t) \in \mathbb{R}^H \,]$ with a shared event loop:

$$\begin{aligned} &z(t_i{+}\Delta) = e^{-B\Delta} z(t_i^+), && z(t_i^+) = z(t_i^-) + w_{k_i}\mathbf{1}, \quad B = \mathrm{diag}(\beta) \text{ fixed},\ w_k = \psi(e_k) \geq 0, \\ &h(t_i{+}\Delta) = e^{-D(h,u)\Delta} h(t_i^+) + \dots, && h(t_i^+) = h(t_i^-) + B_h e_{k_i} \qquad \text{(unconstrained)}. \end{aligned}$$

Readouts respect the lane disciplines: rate $\Lambda = \mu_0 + a^\top z$ ($a \geq 0$, linear, no normalization — certificate lane); marks $p(k|u) = \mathrm{softmax}(\mathrm{MLP}(u))$ (free lane); and the optional *bounded, mean-one gate* coupling the lanes,

$$\tilde{\Lambda}(t) = \Lambda(t)\, g(u(t)), \qquad g = \exp\big(\gamma(\tanh(v^\top u) - \bar{c})\big),\ \gamma = \log G_{\max},\ \bar{c} = \mathrm{EMA}[\tanh(v^\top u)], \quad (20)$$

so $g$ has geometric mean $\approx$1 (pin preserved in expectation; residual absorbed by the verified $\kappa$ calibration) and $\rho_{\mathrm{eff}} \leq G_{\max}^2\, n$ (documented bound). The gate is the *only* path by which the time likelihood reaches the backbone — the fix for LGM's measured underemployment ($\sim$2pp mark gap) — and its $\mathrm{Var}(g)$ contributes bounded *state-dependent* dispersion (regime burstiness) beyond any linear Hawkes, targeting the kurtosis residual. Gateless ($g\equiv$1), the TL-SSM is law-identical to LGM-kf; estimation still splits by lane (ground: Kirchner, convex, frozen; $h$, marks, $g$: MLE).

Verification before launch (all passed): neutral start $g \equiv 1$ exactly (zero-init $v$); observed gate range $[0.50, 2.00]$ within the worst-case bound; time-term gradients reach the backbone decays and impulses; frozen ground untouched by optimization ($n = 0.99$ preserved); geometric mean 0.993 after EMA burn-in. Experiment `lgm-tl` (SOL, 3 seeds, seq 4096 / 48 epochs, fitted ground frozen, $G_{\max} = 2$) ran against `lgm-kf2`; the reads were (a) mark accuracy vs the underemployment gap, (b) $\kappa \approx 1$ retention, (c) Fano-curve retention with kurtosis moving toward real.

**TL-SSM results (SOL; all three seeds, after a calibrator fix).** The initial run exposed a protocol fragility: $\kappa$-bisection assumes the probe rate is a low-noise monotone function of $\kappa$, but near criticality the probe standard error $\sim \sqrt{F \cdot R/T_{\mathrm{total}}}$ exceeds the 5% tolerance and brackets collapse on noise (two of three seeds failed). The repaired calibrator (converged path untouched) adds a replicate-confirmed deep accept and, on bracket exhaustion, a pooled regression on the exact $\kappa$-scaled Hawkes law $1/R(\kappa) = a/\kappa + c$ (linear in $1/\kappa$) with a high-precision confirmation; the full-scale 15% verification remains the protocol gate. All three seeds then complete ($\kappa = 0.982,\ 1.046,\ 0.986$; full-scale rate residuals $-7\%$ to $+13\%$).

The three reads, on full evidence: (a) *Prediction: pass.* Accuracy 0.196–0.197 (kf2 0.194–0.196, SS2P2 0.192), time-NLL $-3.09$ to $-3.16$ (kf2 $-3.035$), time-KS 0.19 (kf2 0.228) — the gate restored the time-likelihood gradient to the backbone. (b) *Pin: fail as an identity* ($\kappa$ spread $[0.982, 1.046]$ vs kf2's $[1.014, 1.019]$), though now operationally robust. (c) *Dispersion: fail on reliability* — and this overturns the single-seed read. Per-seed Fano at $\{1, \dots, 50\}$ s: seed 1 $[18, \dots, 119]$, seed 2 $[46, \dots, 401]$, seed 3 $[54, \dots, 1129]$ against real $[42, \dots, 433]$: right on average, but a $\times 9.5$ spread at 50 s where kf2's transplant determinism holds seed-sd to 2–8%. The learned gate converts dispersion from a set constant into a seed lottery — rate-neutrality's zero-variance transplant guarantee is exactly what the gate spends. Kurtosis improves in mean (38 vs kf2's 22; real 163) but is seed-unstable (13–61). Net: LGM-kf2 remains the shipped simulator; the gate is the right mechanism for prediction and kurtosis but needs its variance *targeted* — smaller $G_{\max}$, a gate log-variance penalty, or dispersion-loss-matched gate training — before it can be promoted.

# 7 The Jain (Compound Hawkes) reference point

Jain's simulator never optimizes or reports the Fano factor; dispersion is scored via volatility signature plots and $|r|$-autocorrelation, and his own signature plots undershoot empirical levels by 2–3× — the published state of the art also misses level while matching shape. What his pipeline gets structurally right:

1. **Estimation by binned-count regression** (Kirchner-style, lin–log lag grid): binning turns the Hawkes process into a vector autoregression on lagged counts whose coefficients are the kernel integrals; the fitting target *is* the cross-scale count autocovariance — the integrand of $F(\Delta)$. Moments, not MLE; the spectrum cannot be misallocated relative to dispersion.
2. Power-law kernels selected by AIC 100% of the time (straight log–log point estimates over six decades of lag, $10^{-4}$–$10^{2}$ s).
3. Kernel-norm matrix eigenvalue capped at 1 by rescaling (his projection); inhibitory cross-kernels retained.
4. Baselines derived, not fitted: $\mu = (\mathbb{I} - M)\Lambda$ — the matrix form of LGM's pin.
5. Compound order sizes (Dirac spikes at round numbers + geometric); the constant-size ablation is wrong by $\sim$100× on price variance. Time-of-day multipliers throughout.

**The RL chapter's acceptance checklist.** His market-making chapter converts stylized facts into functional requirements, ranked by agent ablations: (1) *clustered arrivals as the agent's observable* — removing Hawkes intensities from the RL state flips Sharpe $+31.5 \to -20/-51$, the most catastrophic ablation; (2) *queue dynamics* — removing relative queue position yields pump-and-dump or unprofitable policies; (3) *arbitrage-consistent impact memory* — exponential kernels admit dynamic arbitrage that the optimizer finds ("pump & dump"), rarely so under power-law: mis-specified memory manufactures fake alpha; (4) *responsiveness* — spread widening, queue depletion, liquidity regeneration in reaction to the agent; (5) spread mean-reversion via spread-dependent in-spread intensities ($\propto (s-1)^{0.75}$, $R^2 = 0.85$); (6) realistic frictions (strategies collapse beyond 2 bps). Items (3)–(4) require genuine self-excitation — absent in our measured SS2P2 checkpoints — making the dispersion program a prerequisite for the agent phase, not an aesthetic.

# 8 Conclusions

1. Prediction and simulation are different objectives; one-step MLE is blind to, and given capacity misallocates, the spectrum controlling $F(\Delta)$.

2. The TBPTT gradient horizon must cover the measured scales: the cheapest large win found (BTC 3/3 seeds transformed).
3. Along one training run, dispersion peaks early and accuracy late; no NLL-selected checkpoint dominates.
4. The rate ceiling is the current level bottleneck (a third of events saturated; teacher-forced dispersion nearly right).
5. Certified dispersion requires linear-in-history structure: only LGM has a settable $n$, a pinned mean rate, and closed-form $F(\Delta)$ at any horizon; the Kirchner regression is the proven recipe for fitting its ground (typed, per asset) to cross-scale covariances post hoc, marks untouched.
6. Per-asset targets are mandatory ($n$: 0.948 vs 0.973 is a 4× Fano difference); the tick regime (IS share) should shape typed-kick priors.

**Roadmap.** (1) cap-9 control (in flight); (2) Kirchner binned-regression fit of the LGM ground; (3) LGM ground × bounded neural gate ($\rho_{\text{eff}} \leq n\, G_{\max}$) to restore timing without breaking the certificate; (4) time-of-day multiplier; (5) exploitability audit before trusting any policy trained in the simulator.

*Provenance: experiments under* `peacock:~/simulation/experiments/ma_cbse/`*; code* `honglinfu98/simulation` *commits* `7b97b21..88d22cc`*; full lab log in* `docs/fano_investigation_2026-08.md` *of this repository.*